Software Requirements Specification

SRS v1.1 — Detailed Use Case Catalogue

Wholesale Stationery, Notebook & FMCG Business Management Platform

Baseline: SRS v1.0   **Business:** Tumkur, Karnataka, India   **Date:** 30 September 2026

Purpose: convert the SRS v1.0 requirements into testable, end-to-end operational use cases. This document does not change the fundamental scope established by v1.0; it makes the four priority workflows explicit enough to guide architecture, UX, APIs, database design and testing.

1. Scope and Conventions

SRS v1.1 covers four business workflows and their supporting use cases. Each use case identifies the actor, trigger, preconditions, inputs, main flow, alternate flows, exceptions, business rules, outputs, data effects, permissions, audit events and acceptance criteria.

1.1 Core Workflow Chain

Purchase: Supplier → Purchase Order → Goods Receipt → Inventory → Supplier Invoice → Payable → Payment

Sales: Customer → Cart/Order → Pricing → Tax → Invoice → Payment/Credit → Inventory → Receivable

School: School Requirement → Quotation → Customer PO → Fulfilment/Delivery → Invoice → Payment

Accounting: Source Transaction → Journal/Account Effect → Receivable/Payable/Cash/Bank/Expense → Period Reporting

1.2 Transaction Integrity Principle

A confirmed business transaction must produce the appropriate inventory and/or accounting effects exactly once. Corrections should be represented by controlled reversal, return, adjustment or cancellation transactions rather than silently overwriting history.

2. Actors

• **Owner/Admin:** unrestricted authorized management and approval role.

• **Manager:** operational approvals and management functions within assigned permissions.

• **Accountant:** accounting, receipts, payments, expenses and financial reports.

• **Sales Staff:** customer orders, quotations and sales within assigned limits.

• **Billing Staff:** billing/POS and payment collection.

• **Warehouse Staff:** receiving, stock movement and delivery/dispatch operations.

• **Supplier:** external party providing goods.

• **Customer:** retail, wholesale/B2B, school, office or other institution.

• **System:** performs validation, calculation, persistence, inventory posting, accounting posting and notifications.

3. Common Transaction States

Entity	Typical states	Key rule
Purchase Order	**Draft → Approved → Partially Received → Fully Received → Closed / Cancelled**	**Approval and receipt are distinct.**
Goods Receipt	**Draft → Confirmed → Reversed**	**Only confirmed receipt posts stock.**
Quotation	**Draft → Sent → Accepted / Rejected / Expired / Cancelled**	**Acceptance does not itself create stock movement.**
Sales Order	**Draft → Confirmed → Partially Fulfilled → Fulfilled / Cancelled**	**Stock commitment and actual stock issue are distinct if implemented.**
Invoice	**Draft → Posted → Partially Paid → Paid / Cancelled / Reversed**	**Posted invoice is auditable.**
Payment	**Draft → Posted → Reversed**	**Posted payment affects ledger.**

4. Workflow A — Purchase → Inventory

Objective: acquire goods from suppliers, receive them accurately, update stock, create the payable and record settlement.

UC-PUR-001 — Create / Maintain Supplier

Priority	Must

Primary actor(s)	Owner/Admin, Manager
Trigger	A new supplier is identified or supplier master data changes.
Preconditions	User has supplier-management permission.
Inputs	Supplier identity, contact, GST/tax details where applicable, payment terms, address and status.

Main Flow

1. User opens supplier management.

2. System validates required fields and duplicate-risk fields.

3. User saves supplier.

4. System assigns supplier identifier and audit metadata.

5. Supplier becomes available for purchasing and accounting.

Alternate Flows

1. Existing supplier is updated rather than duplicated.

2. Supplier can be marked inactive instead of deleted when historical transactions exist.

Exceptions / Failure Handling

• Missing mandatory data; invalid GST information format where validation is implemented; suspected duplicate supplier; unauthorized user.

Business Rules

• Supplier identifiers must remain stable after transactions exist.

• Historical transactions must retain their supplier reference.

Outputs

• Supplier master record; supplier ID; status.

Data Changes / Transaction Effects

• Create/update supplier master; no inventory effect; no accounting posting from master creation alone.

Permissions

• Create/update supplier; deactivate supplier; view supplier financial history.

Audit Events

• Supplier creation/update/deactivation; before/after sensitive fields; actor and timestamp.

Acceptance Criteria

AC-001-01: Supplier can be created with required data.

AC-001-02: Duplicate supplier warning is produced where matching rules detect likely duplication.

AC-001-03: Historical supplier transactions remain accessible after deactivation.

UC-PUR-002 — Create Purchase Order

Priority	Must
Primary actor(s)	Purchase Staff / Manager
Trigger	Business decides to purchase stock from a supplier.
Preconditions	Supplier exists and is active; products exist and are active; user has purchasing permission.
Inputs	Supplier, store, products, quantities, UOM, agreed purchase price, expected date, taxes/terms and notes.

Main Flow

1. User selects supplier and store.

2. User adds products and quantities using configured UOM.

3. System validates UOM conversions and product status.

4. System calculates line values and applicable tax information.

5. User saves draft and submits for approval where required.

6. Approved PO receives a unique number and status.

7. PO is available for receipt and supplier reference.

Alternate Flows

1. Partial order can be created.

2. User may revise draft before approval.

3. Authorized manager may reject/cancel an unreceived PO.

Exceptions / Failure Handling

• Product inactive; conversion missing; invalid quantity; price outside user authority; supplier inactive; duplicate PO reference where uniqueness is required.

Business Rules

• A PO is a commitment, not stock.

• Creating a PO must not increase inventory.

• Approved PO should retain the agreed commercial terms.

Outputs

• Purchase Order number, status, totals, expected receipt date and supplier reference.

Data Changes / Transaction Effects

• PO header/lines created; no stock posting; optional accounting commitment only if later designed.

Permissions

• Create/edit draft; submit/approve/reject/cancel according to role.

Audit Events

• Create, edit, approve, reject, cancel and price override events.

Acceptance Criteria

AC-002-01: Draft PO can be saved and resumed.

AC-002-02: Approved PO has immutable or controlled commercial history.

AC-002-03: Inventory remains unchanged until goods receipt.

UC-PUR-003 — Receive Goods

Priority	Must
Primary actor(s)	Warehouse Staff, Manager
Trigger	Supplier delivers goods against a purchase order.
Preconditions	PO exists and is receivable; receiving user has permission; products and UOMs are valid.
Inputs	PO, received quantities, UOM, batch/lot, expiry where required, condition, actual receipt date, supplier document reference.

Main Flow

1. User selects PO.

2. System displays ordered quantities.

3. User enters actual received quantities.

4. System validates received quantity against permitted over/under-receipt rules.

5. User records batch/expiry for configured products.

6. User records damaged/short/excess quantities as appropriate.

7. User confirms goods receipt.

8. System creates stock-in movements for accepted goods.

9. System updates PO status based on remaining quantity.

Alternate Flows

1. Partial receipt.

2. Over-receipt can be rejected, accepted with authorization, or placed on hold according to future configuration.

3. Damaged goods may be recorded separately from accepted stock.

Exceptions / Failure Handling

• Quantity mismatch; invalid batch; missing expiry for required product; duplicate receipt; PO already closed; stock posting failure.

Business Rules

• Only confirmed receipt increases inventory.

• Each receipt must be linked to its PO and supplier.

• Batch-controlled products must carry batch data.

• Received quantity must not be posted twice.

Outputs

• Goods Receipt number; accepted quantity; rejected/damaged quantity; updated PO status; stock ledger entries.

Data Changes / Transaction Effects

• Inventory increases; batch stock created/updated; PO received quantity updated; supplier receipt reference stored.

Permissions

• Receive goods; edit draft receipt; confirm; reverse if authorized.

Audit Events

• Receipt creation/confirmation/reversal; quantity changes; batch/expiry changes; actor/time.

Acceptance Criteria

AC-003-01: Accepted quantity increases stock exactly once.

AC-003-02: Partial receipt leaves remaining PO quantity visible.

AC-003-03: Batch-controlled products cannot be confirmed without required batch data.

AC-003-04: Receipt is traceable to the source PO.

UC-PUR-004 — Post Supplier Invoice / Payable

Priority	Must
Primary actor(s)	Accountant, Manager
Trigger	Supplier invoice is received and matched to procurement.
Preconditions	Goods receipt exists or business policy permits invoice-before-receipt; supplier exists; user has accounting permission.
Inputs	Supplier invoice number/date, receipt/PO reference, taxable value, GST, total, due date, payment terms and accounting allocation.

Main Flow

1. User selects supplier and relevant receipt/PO.
2. System compares quantities and amounts against source transactions.
3. User enters invoice/tax details.
4. System validates duplicate supplier invoice references.
5. User posts invoice.
6. System creates accounts payable and accounting entries.
7. Invoice becomes available for payment.

Alternate Flows

1. Invoice may cover multiple receipts.
2. Differences may be placed on hold for review.
3. Authorized user can record an adjustment with reason.

Exceptions / Failure Handling

• Duplicate invoice; mismatch beyond tolerance; invalid tax data; missing source; closed accounting period.

Business Rules

• Supplier invoice posting creates payable but does not duplicate inventory already posted by receipt.

• Invoice cancellation/reversal must reverse accounting effect.

Outputs

• Supplier payable invoice; due date; outstanding amount; accounting journal.

Data Changes / Transaction Effects

• Accounts payable increases; supplier balance increases; no duplicate inventory movement.

Permissions

• Create/post/reverse supplier invoice according to accounting permissions.

Audit Events

• Invoice posting/reversal; tax edits; adjustments; approvals.

Acceptance Criteria

AC-004-01: Duplicate invoice numbers are detected according to configured supplier scope.

AC-004-02: Posted invoice appears in supplier outstanding.

AC-004-03: Accounting entry balances.

UC-PUR-005 — Pay Supplier

Priority	Must
Primary actor(s)	Accountant, Owner/Admin
Trigger	Supplier payment is due or approved for settlement.
Preconditions	Posted supplier payable exists; payment account is available; user has payment permission.
Inputs	Supplier, invoices/credits, payment amount, payment date, payment method, bank/cash account, reference.

Main Flow

1. User selects supplier and outstanding invoices.
2. System shows outstanding payable.
3. User enters payment amount and method.
4. System validates available/allowed payment account and allocation.

5. User posts payment.

6. System reduces payable and updates supplier balance.

7. Receipt/reference is stored.

Alternate Flows

1. Partial payment.

2. One payment can be allocated across multiple invoices.

3. Unallocated supplier advance may be supported if accounting design permits.

Exceptions / Failure Handling

• Overpayment; insufficient account balance if controlled; duplicate payment reference; closed period; unauthorized user.

Business Rules

• Posted payment cannot silently disappear; reversal is a separate controlled action.

• Allocation must not exceed allowable outstanding unless advance is explicitly supported.

Outputs

• Payment transaction, allocations, supplier outstanding and account balance.

Data Changes / Transaction Effects

• Accounts payable decreases; cash/bank decreases; supplier balance updated.

Permissions

• Create/post/reverse payment; view supplier payable.

Audit Events

• Payment posting/reversal; allocation changes; payment method/account.

Acceptance Criteria

AC-005-01: Full and partial supplier payments are correctly reflected.

AC-005-02: Supplier outstanding reconciles to invoice less posted payments/credits.

AC-005-03: Payment is auditable.

5. Workflow B — Retail/Wholesale Sale → Invoice → Payment → Inventory

Objective: sell products through retail or wholesale channels, apply the correct price and GST, issue an invoice, collect payment or create credit, and reduce inventory exactly once.

UC-SAL-001 — Create Customer / Configure Customer Pricing

Priority	Must
Primary actor(s)	Sales Staff, Manager, Owner/Admin
Trigger	A new customer is identified or commercial terms change.
Preconditions	User has customer-management permission.
Inputs	Customer type, identity, contact, GSTIN where applicable, address, credit limit, payment terms, pricing group and customer-specific prices.

Main Flow

1. User creates or opens customer.

2. System validates required information.

3. User configures permitted pricing and credit settings.

4. System saves customer and effective-date metadata.

5. Customer becomes selectable in sales.

Alternate Flows

1. Walk-in retail sale can use a generic customer profile if policy permits.

2. Specific negotiated price can be added without changing the global product price.

Exceptions / Failure Handling

• Duplicate customer; invalid GSTIN; price below authorized threshold; credit limit above user authority.

Business Rules

• Customer-specific price rules must be auditable and should have clear effective dates.

• Customer profile changes must not rewrite historical invoice prices.

Outputs

• Customer profile; pricing rules; credit terms.

Data Changes / Transaction Effects

• Customer master updated; no inventory/accounting effect until a transaction occurs.

Permissions

• Create/edit/deactivate customer; configure pricing; configure credit terms.

Audit Events

• Customer changes; price-rule changes; credit-limit changes.

Acceptance Criteria

AC-001-01: Historical invoices retain original prices.

AC-001-02: Unauthorized users cannot create unrestricted price overrides.

AC-001-03: Customer is available for authorized sales.

UC-SAL-002 — Create Retail / Wholesale Cart or Sales Order

Priority	Must
Primary actor(s)	Billing Staff, Sales Staff
Trigger	Customer requests products or a walk-in sale begins.
Preconditions	Products are active; user has sales permission; pricing rules are available.
Inputs	Customer/generic customer, products/barcodes, quantities/UOM, discounts, delivery/store context.

Main Flow

1. User scans barcode or searches product.

2. System identifies product and available stock.

3. System determines applicable price based on customer, quantity and pricing rules.

4. User enters quantity and permitted discount.

5. System calculates line totals and estimated tax.

6. User reviews cart/order.

7. System saves transaction as draft/confirmed according to channel.

Alternate Flows

1. Retail walk-in may proceed directly to invoice.

2. Wholesale order may be saved before billing.

3. Product can be added manually when no barcode exists.

4. Backordered/insufficient stock may be allowed only if business policy supports it.

Exceptions / Failure Handling

• Unknown barcode; inactive product; insufficient stock; invalid UOM; unauthorized discount; price below floor; credit limit exceeded.

Business Rules

• Price selection must be deterministic and auditable.

• A draft cart does not reduce physical inventory unless an explicit reservation mechanism is later designed.

• Wholesale and retail may use different pricing.

Outputs

• Cart/order; line pricing; discounts; estimated taxes; availability warnings.

Data Changes / Transaction Effects

• Sales order/cart created; no final stock reduction until confirmation/fulfilment.

Permissions

• Create/edit cart/order; discount within limit; price override with approval.

Audit Events

• Price override; discount; stock warning; order confirmation/cancellation.

Acceptance Criteria

AC-002-01: Correct price is selected for customer.

AC-002-02: Barcode identifies correct product.

AC-002-03: Unauthorized price/discount changes are blocked.

AC-002-04: Draft does not unexpectedly reduce stock.

UC-SAL-003 — Generate and Post GST Invoice

Priority	Must
Primary actor(s)	Billing Staff, Manager
Trigger	Sale is ready for billing.
Preconditions	Sale/cart is valid; customer/tax data available; products have tax configuration; invoice number can be assigned.
Inputs	Customer, line items, prices, discounts, HSN/SAC, GST rate, place-of-supply/tax context as required, payment/credit terms.

Main Flow

1. System validates sale lines and tax configuration.

2. System calculates taxable value and applicable GST components.

3. System generates invoice number according to configured series.

4. User reviews invoice.

5. Authorized user posts invoice.

6. System stores immutable/auditable invoice snapshot.

7. System creates receivable or payment settlement path.

Alternate Flows

1. Tax-inclusive prices can be converted to taxable value using configured rules.

2. Retail invoice may use walk-in customer data where legally/business appropriate.

3. Invoice can be printed or sent through WhatsApp integration.

Exceptions / Failure Handling

• Tax configuration missing; invoice numbering conflict; invalid customer tax data; closed accounting period; calculation mismatch.

Business Rules

• Posted invoice must preserve the exact commercial/tax snapshot used at posting.

• Changing product master later must not alter historical invoices.

• Invoice cancellation/reversal is controlled.

Outputs

• Posted GST invoice; invoice number; tax breakdown; receivable/payment status.

Data Changes / Transaction Effects

• Sales invoice recorded; revenue/tax/receivable accounting entries created; stock reduction occurs according to configured sale-posting point.

Permissions

• Post invoice; cancel/reverse according to authority; print; send through WhatsApp.

Audit Events

• Invoice creation/posting; tax overrides; cancellation/reversal; print/send event.

Acceptance Criteria

AC-003-01: Invoice totals reconcile line values + tax.

AC-003-02: Correct GST components are shown.

AC-003-03: Historical invoice remains unchanged after product master changes.

AC-003-04: Invoice is traceable to source sale.

UC-SAL-004 — Collect Customer Payment / Create Credit

Priority	Must
Primary actor(s)	Billing Staff, Accountant, Manager
Trigger	Customer pays at sale or invoice is sold on credit.
Preconditions	Posted invoice exists; payment method is configured; user has payment permission.
Inputs	Amount, payment method, cash/bank/UPI/card reference as applicable, allocation, date.

Main Flow

1. System shows invoice amount and outstanding.

2. User selects payment method.

3. User enters amount/reference.

4. System validates amount and allocation.

5. User posts payment.

6. System updates invoice status and customer outstanding.

7. Receipt/reference is stored.

Alternate Flows

1. Full payment.

2. Partial payment.

3. Credit sale with zero immediate payment.

4. Advance/unallocated payment if enabled by accounting design.

Exceptions / Failure Handling

• Overpayment without permitted advance; duplicate transaction reference; invalid payment account; closed period.

Business Rules

• Payment posting must reduce receivable exactly once.

• A credit sale increases outstanding; immediate payment reduces it.

Outputs

• Receipt/payment transaction; invoice balance; customer outstanding; payment status.

Data Changes / Transaction Effects

• Accounts receivable decreases; cash/bank increases; customer balance updated.

Permissions

• Post/reverse payment; view customer outstanding.

Audit Events

• Payment posting/reversal; allocation changes; payment method/reference.

Acceptance Criteria

AC-004-01: Full payment marks invoice paid.

AC-004-02: Partial payment leaves correct outstanding.

AC-004-03: Customer balance reconciles to invoices less payments/credits.

UC-SAL-005 — Post Inventory Reduction / Fulfilment

Priority	Must
Primary actor(s)	Billing Staff, Warehouse Staff, Manager
Trigger	Confirmed sale is ready for stock issue/fulfilment.
Preconditions	Sale/invoice is confirmed; stock is available or approved exception exists; warehouse/store context is known.
Inputs	Sale lines, quantities/UOM, batch selection where applicable, fulfilment/dispatch information.

Main Flow

1. System determines stock to issue.

2. For batch-controlled products, system selects or requests batch.

3. System validates available stock.

4. Authorized user confirms stock issue.

5. System posts inventory-out transactions.

6. System links stock movements to sale/invoice.

Alternate Flows

1. Multiple batches may satisfy a line.

2. Partial fulfilment can be supported for wholesale orders.

3. Authorized stock adjustment may be used for approved discrepancy, not to hide a failed sale.

Exceptions / Failure Handling

• Insufficient stock; batch expired; batch not found; duplicate posting; warehouse/store mismatch.

Business Rules

• One sale must not reduce stock twice.

• Batch-controlled products should use valid/non-expired stock according to configured policy.

• Inventory movement must reference the source sale.

Outputs

• Stock issue; remaining fulfilment quantity; batch depletion.

Data Changes / Transaction Effects

• Inventory decreases; batch quantity decreases; sale fulfilment status updated.

Permissions

• Confirm/reverse stock issue; select batch; partial fulfilment.

Audit Events

• Stock issue/reversal; batch selection; manual override; discrepancy.

Acceptance Criteria

AC-005-01: Inventory reduces exactly once.

AC-005-02: Stock cannot go negative unless explicit negative-stock policy is later approved.

AC-005-03: Batch balances remain correct.

AC-005-04: Sale and inventory ledger are linked.

UC-SAL-006 — Sales Return / Cancellation / Reversal

Priority	Must
Primary actor(s)	Manager, Owner/Admin, Accountant
Trigger	Customer returns goods or a posted sale must be corrected.
Preconditions	Original sale/invoice exists; user has reversal/return permission; return quantity is within policy.
Inputs	Original invoice, returned products/quantity, reason, condition, batch, refund/credit method.

Main Flow

1. User selects original invoice.

2. System shows eligible quantities.

3. User records returned quantity and reason.

4. System determines whether stock is saleable, damaged or expired.

5. System creates return/reversal transaction.

6. Accounting and tax effects are reversed/adjusted.

7. Customer balance/refund is updated.

Alternate Flows

1. Partial return.

2. Return to stock as saleable.

3. Return to damaged/quarantine stock.

4. Credit note/refund according to business/accounting policy.

Exceptions / Failure Handling

• Return exceeds sold quantity; unknown batch; invoice already fully reversed; closed period; refund account unavailable.

Business Rules

• Returns must reference the original sale.

• Return must not erase the original invoice.

• Tax/accounting reversal must match the return.

Outputs

• Return document/credit note; stock return; customer balance/refund.

Data Changes / Transaction Effects

• Inventory increases only for saleable returned goods; damaged stock tracked separately; revenue/tax/receivable effects reversed as applicable.

Permissions

• Create/approve return; reverse invoice; issue credit/refund.

Audit Events

• Return/reversal; quantity; batch; reason; refund/credit.

Acceptance Criteria

AC-006-01: Returned quantity cannot exceed eligible quantity.

AC-006-02: Original transaction remains auditable.

AC-006-03: Inventory and accounting effects reconcile.

6. Workflow C — School Requirement → Quotation → PO → Delivery → Invoice → Payment

Objective: support school/institutional business as a distinct workflow because purchases are often periodic, bulk, negotiated and based on quotations/purchase orders.

UC-SCH-001 — Capture School Requirement

Priority	Must
Primary actor(s)	Sales Staff, Manager
Trigger	School communicates a stationery/product requirement.
Preconditions	School exists as a customer; user has school-sales permission.
Inputs	School, academic/period context, requirement date, requested products, quantities, preferred specifications, target date and notes.

Main Flow

1. User opens school requirement.

2. User selects products or records requested specifications.

3. System validates product availability/master data where possible.

4. User records requested quantities and target dates.

5. Requirement is saved.

6. System makes requirement available for quotation.

Alternate Flows

1. Requirement can be entered in stages.

2. Products not yet in catalogue can be marked as pending master-data creation rather than silently substituted.

3. Historical requirements remain available.

Exceptions / Failure Handling

• Ambiguous product specification; duplicate requirement; invalid quantity; inactive school.

Business Rules

• Requirement is not a confirmed sale and must not reduce inventory or create revenue.

• One requirement may generate one or multiple quotations/orders.

Outputs

• Requirement record; requirement lines; status.

Data Changes / Transaction Effects

• No stock/accounting effect.

Permissions

• Create/edit/close requirement; convert to quotation.

Audit Events

• Create/update/close; product substitutions; quantity changes.

Acceptance Criteria

AC-001-01: Requirement retains original requested quantities.

AC-001-02: No inventory movement occurs.

AC-001-03: Requirement can be converted to quotation without rekeying data unnecessarily.

UC-SCH-002 — Prepare and Send School Quotation

Priority	Must
Primary actor(s)	Sales Staff, Manager
Trigger	School requirement is ready for commercial quotation.
Preconditions	Requirement exists; pricing is available; user has quotation permission.
Inputs	Requirement, quoted quantities, prices, discounts, GST, validity, delivery terms and notes.

Main Flow

1. System copies requirement lines into quotation.

2. User applies school/customer pricing and authorized discounts.

3. System calculates taxes and totals.

4. User sets validity and commercial terms.

5. Quotation is reviewed/approved where required.

6. System assigns quotation number and status.

7. Quotation is delivered/printed/shared.

Alternate Flows

1. Quotation can contain alternatives.

2. Multiple quotations may be created for revised requirements.

3. Expired quotation can be recreated or revised with new version/history.

Exceptions / Failure Handling

• Unauthorized discount; tax error; missing price; quotation expiry conflict.

Business Rules

• Quotation is not an invoice and does not reduce stock.

• Commercial terms should be versioned/auditable.

• Acceptance must refer to a specific quotation version.

Outputs

• Quotation document, totals, validity, status.

Data Changes / Transaction Effects

• Quotation master/lines; no stock movement; no revenue posting.

Permissions

• Create/edit/approve/send/expire/cancel quotation.

Audit Events

• Price override, discount, approval, send and version events.

Acceptance Criteria

AC-002-01: Quotation totals calculate correctly.

AC-002-02: School receives a stable quotation reference/version.

AC-002-03: Quotation does not alter stock or accounting.

UC-SCH-003 — Record School Purchase Order / Acceptance

Priority	Must
Primary actor(s)	Sales Staff, Manager
Trigger	School accepts quotation and provides a PO/order confirmation.
Preconditions	Quotation is valid or an authorized exception is recorded; school exists; user has order permission.
Inputs	Quotation reference, school PO number/date, accepted quantities, agreed price changes, delivery schedule and terms.

Main Flow

1. User selects accepted quotation.
2. System copies agreed lines and prices.
3. User records school PO number/date.
4. System validates any deviations.
5. Authorized user confirms order.
6. System changes order status and makes it available for fulfilment.

Alternate Flows

1. School may accept only part of quotation.
2. PO may include additional or reduced quantities subject to authorization.
3. PO may be entered manually if quotation was external.

Exceptions / Failure Handling

• Quotation expired; price mismatch; quantity mismatch; duplicate school PO; unauthorized change.

Business Rules

• Customer PO acceptance creates an order commitment, not physical stock movement.

• Any deviation from quotation must be recorded.

Outputs

• Sales order/PO reference; accepted lines; delivery schedule.

Data Changes / Transaction Effects

• Order created; inventory may be reserved only if reservation is explicitly designed; no physical stock issue yet.

Permissions

• Create/confirm/amend/cancel school order.

Audit Events

• PO creation, deviations, approvals and cancellation.

Acceptance Criteria

AC-003-01: Accepted quantities and prices are preserved.

AC-003-02: School PO reference is unique within configured scope.

AC-003-03: No physical stock reduction occurs at acceptance.

UC-SCH-004 — Fulfil and Deliver School Order

Priority	Must
Primary actor(s)	Warehouse Staff, Sales Staff, Manager
Trigger	Confirmed school order is scheduled for fulfilment.
Preconditions	Order confirmed; stock available or approved procurement/backorder exists; delivery details known.
Inputs	Order lines, quantities, batches where applicable, delivery address, dispatch/delivery reference and recipient confirmation.

Main Flow

1. System shows fulfilment quantities.

2. Warehouse picks products.

3. System records batch where applicable.

4. User confirms dispatch/stock issue.

5. System reduces inventory.

6. Delivery status is updated.

7. Delivery confirmation/reference is stored.

Alternate Flows

1. Partial delivery.

2. Backorder remaining quantity.

3. Multiple delivery documents for one order.

4. Customer pickup instead of delivery if supported.

Exceptions / Failure Handling

• Insufficient stock; wrong product; expired batch; duplicate dispatch; delivery discrepancy.

Business Rules

• Stock reduction occurs at confirmed stock issue/dispatch according to configured policy.

• Partial delivery must leave remaining quantity visible.

• Delivered quantity cannot exceed ordered quantity without authorization.

Outputs

• Delivery/dispatch record; fulfilled quantity; remaining quantity; stock movements.

Data Changes / Transaction Effects

• Inventory decreases; order fulfilment status changes.

Permissions

• Pick, dispatch, confirm delivery, reverse authorized stock issue.

Audit Events

• Picking, batch selection, dispatch, delivery confirmation and reversal.

Acceptance Criteria

AC-004-01: Delivered quantity is traceable to order lines.

AC-004-02: Partial delivery is correctly represented.

AC-004-03: Inventory matches dispatched quantity.

UC-SCH-005 — Invoice School Order

Priority	Must
Primary actor(s)	Billing Staff, Accountant, Manager
Trigger	Goods are delivered or invoice trigger condition is met.
Preconditions	Order/fulfilment meets invoicing policy; customer tax data is valid; invoice permissions exist.
Inputs	Order/delivery references, quantities, prices, discounts, HSN/SAC, GST and payment terms.

Main Flow

1. User selects fulfilled order/delivery.

2. System copies delivered/billable quantities.

3. System calculates taxes and totals.

4. Invoice is reviewed and posted.

5. System creates receivable.

6. Invoice is printable and eligible for WhatsApp delivery.

Alternate Flows

1. Partial invoice for partial delivery.

2. Invoice may be generated at order stage if business policy permits for a specific contract.

Exceptions / Failure Handling

• Quantity mismatch; missing tax data; duplicate invoice; closed period.

Business Rules

• Invoice quantity must not exceed billable quantity unless authorized.

• Historical invoice snapshot is immutable/auditable.

Outputs

• GST invoice; receivable; invoice status.

Data Changes / Transaction Effects

• Revenue/tax/receivable posted; no duplicate stock movement.

Permissions

• Create/post/reverse school invoice; print/send.

Audit Events

• Invoice posting, tax override, reversal and delivery linkage.

Acceptance Criteria

AC-005-01: Invoice equals billable delivered/approved quantity.

AC-005-02: Receivable is created correctly.

AC-005-03: Invoice is linked to school order and delivery.

UC-SCH-006 — Receive School Payment

Priority	Must
Primary actor(s)	Accountant, Manager
Trigger	School pays invoice or account balance.
Preconditions	Posted school invoice exists; payment method/account is configured.
Inputs	Payment amount, method, reference, invoice allocation and date.

Main Flow

1. User selects school outstanding.
2. System displays invoices and balances.
3. User records payment.
4. System validates and posts payment.
5. Invoice status and school balance are updated.
6. Receipt/reference is retained.

Alternate Flows

1. Partial payment.
2. One payment allocated across multiple invoices.
3. Advance payment if supported by accounting design.

Exceptions / Failure Handling

• Overpayment, duplicate reference, invalid account, closed period.

Business Rules

• Payment cannot reduce receivable below zero unless advance/credit handling explicitly supports it.

Outputs

• Payment receipt; invoice balance; school outstanding.

Data Changes / Transaction Effects

• Receivable decreases; cash/bank increases.

Permissions

• Post/reverse payment; allocate payment.

Audit Events

• Payment posting/reversal/allocation.

Acceptance Criteria

AC-006-01: School outstanding reconciles to invoices less payments/credits.

AC-006-02: Partial payment remains correctly outstanding.

7. Workflow D — Accounting → Customer/Supplier/Cash/Bank/Expense → Financial Reports

Objective: maintain a complete accounting trail from operational transactions and direct financial transactions through financial statements and management reports.

UC-ACC-001 — Post Customer Receipt

Priority	Must
Primary actor(s)	Accountant, Manager

Trigger	Customer payment is received.
Preconditions	Customer receivable exists or payment is explicitly recorded as an advance; user has accounting permission.
Inputs	Customer, invoice allocation, amount, payment method, cash/bank account, date, reference.

Main Flow

1. Select customer and outstanding.

2. Enter payment.

3. Allocate to one or more invoices.

4. Validate amount and account.

5. Post receipt.

6. Update receivable and cash/bank ledger.

7. Generate receipt/reference.

Alternate Flows

1. Partial payment; multiple invoices; advance.

Exceptions / Failure Handling

• Invalid allocation; duplicate reference; closed period; unauthorized account.

Business Rules

• Every posted receipt must have balanced accounting effect.

• Receipt reversal is controlled.

Outputs

• Receipt, allocations, customer balance, cash/bank balance.

Data Changes / Transaction Effects

• Dr Cash/Bank; Cr Receivable or Advance as applicable.

Permissions

• Post/reverse; allocate.

Audit Events

• Receipt creation, posting, allocation and reversal.

Acceptance Criteria

AC-001-01: Customer balance decreases correctly.

AC-001-02: Cash/bank and receivable ledgers balance.

UC-ACC-002 — Post Supplier Payment

Priority	Must
Primary actor(s)	Accountant, Manager
Trigger	Supplier payment is made.
Preconditions	Supplier payable exists or advance is explicitly allowed.
Inputs	Supplier, invoices, amount, payment method/account, date, reference.

Main Flow

1. Select supplier payable.

2. Enter amount and allocation.

3. Validate.

4. Post payment.

5. Update payable and cash/bank.

6. Store reference.

Alternate Flows

1. Partial payment; multiple invoices; advance.

Exceptions / Failure Handling

• Overpayment; duplicate reference; closed period; invalid account.

Business Rules

• Posted supplier payment reduces payable exactly once.

Outputs

• Payment, allocation, supplier balance, cash/bank balance.

Data Changes / Transaction Effects

• Dr Payable; Cr Cash/Bank, subject to accounting configuration.

Permissions

• Post/reverse/allocate.

Audit Events

• Posting, allocation, reversal.

Acceptance Criteria

AC-002-01: Supplier balance reconciles to invoices/payments/credits.

AC-002-02: Cash/bank ledger reflects payment.

UC-ACC-003 — Record Expense

Priority	Must
Primary actor(s)	Accountant, Manager
Trigger	Business incurs an operating or other expense.
Preconditions	Expense category/account exists; payment source exists; user has expense permission.
Inputs	Expense date, vendor/payee, category/account, amount, tax where applicable, payment source, reference, attachment if supported.

Main Flow

1. User selects expense account/category.

2. Enters amount and payment source.

3. Adds tax details if applicable.

4. System validates accounting period and required fields.

5. User posts expense.

6. System updates expense and cash/bank payable effect.

Alternate Flows

1. Expense can be paid immediately or recorded as payable if accounting model supports it.

2. Recurring expense templates may be future functionality.

Exceptions / Failure Handling

• Invalid account; closed period; missing amount; duplicate document.

Business Rules

• Expense must be traceable to source document/reference where applicable.

• Changing an expense after posting requires controlled correction/reversal.

Outputs

• Expense transaction; account balances; cash/bank or payable effect.

Data Changes / Transaction Effects

• Expense account increases; cash/bank decreases or payable increases.

Permissions

• Create/post/reverse expense.

Audit Events

• Expense creation, edits before posting, posting and reversal.

Acceptance Criteria

AC-003-01: Expense appears in P&L under correct account.

AC-003-02: Payment source balance is correct.

AC-003-03: Posted expense is auditable.

UC-ACC-004 — Record Cash Transaction

Priority	Must
Primary actor(s)	Accountant, Manager
Trigger	Cash is received, paid, transferred or adjusted through an approved process.
Preconditions	Cash account exists; user has permission.
Inputs	Date, transaction type, amount, counterparty, reference, account allocation.

Main Flow

1. User selects cash account.

2. Records transaction type and amount.

3. System validates balancing account.

4. User posts transaction.

5. Cash balance updates.

6. Reference is stored.

Alternate Flows

1. Cash transfer between internal accounts may be represented as paired entries.

2. Opening cash balance can be recorded during setup.

Exceptions / Failure Handling

• Negative cash if prohibited; invalid account; duplicate transaction.

Business Rules

• Cash should reconcile to posted cash transactions and approved opening/closing balances.

Outputs

• Cash ledger, balance, source reference.

Data Changes / Transaction Effects

• Cash account increases/decreases according to transaction.

Permissions

• Post/reverse; reconcile.

Audit Events

• Cash transaction and reversal.

Acceptance Criteria

AC-004-01: Cash balance equals opening balance plus posted movements.

AC-004-02: Reversal is traceable.

UC-ACC-005 — Record Bank Transaction

Priority	Must
Primary actor(s)	Accountant, Manager
Trigger	Bank transaction occurs or bank statement is reconciled.
Preconditions	Bank account exists; user has permission.
Inputs	Bank account, date, amount, type, reference, counterparty, related source transaction.

Main Flow

1. User records or imports bank transaction if integration exists later.

2. System matches to known transaction where possible.

3. User confirms unmatched transaction classification.

4. System posts accounting entry.

5. Bank balance updates.

Alternate Flows

1. One bank transaction can be matched to multiple business transactions.

2. Unmatched transaction can be held for review.

Exceptions / Failure Handling

• Duplicate transaction; incorrect bank account; closed period.

Business Rules

• Bank reconciliation must not silently duplicate customer/supplier payments already posted.

Outputs

• Bank ledger, reconciliation status, account balance.

Data Changes / Transaction Effects

• Bank account increases/decreases; related receivable/payable/expense/etc. account changes.

Permissions

• Record, match, classify, reconcile, reverse.

Audit Events

• Bank transaction creation, match, classification and reversal.

Acceptance Criteria

AC-005-01: Bank balance reconciles to recorded transactions.

AC-005-02: Duplicate matches are prevented or flagged.

UC-ACC-006 — Post Sales / Purchase Accounting Entries

Priority	Must
Primary actor(s)	System, Accountant
Trigger	A sales invoice, purchase invoice, credit note, return or other source transaction is posted.
Preconditions	Source transaction is valid and within an open accounting period.
Inputs	Source document, accounts, taxable amount, tax amount, total, date, customer/supplier.

Main Flow

1. System derives journal entries from configured posting rules.

2. System validates debit/credit balance.

3. System posts journal linked to source document.

4. Source transaction status becomes accounting-posted.

Alternate Flows

1. Correction is handled by reversal and reposting where allowed.

2. Manual journal adjustment may be used only by authorized accountant.

Exceptions / Failure Handling

• Unconfigured account mapping; unbalanced journal; closed period; duplicate posting.

Business Rules

• Every source transaction should have at most one active posting set unless explicitly designed for adjustments.

• Journal must balance.

Outputs

• Journal entry, source reference, account balances.

Data Changes / Transaction Effects

• Revenue/COGS/tax/receivable/payable/inventory effects according to accounting design.

Permissions

• View/post/reverse; mapping administration.

Audit Events

• Automatic posting, reversal, mapping change and manual override.

Acceptance Criteria

AC-006-01: Debits equal credits.

AC-006-02: Source transaction and journal are linked.

AC-006-03: Duplicate posting is prevented.

UC-ACC-007 — Period Close / Reopen Control

Priority	Must
Primary actor(s)	Owner/Admin, Accountant
Trigger	Accounting period is ready for close.
Preconditions	Required reconciliations and approvals are complete; user has close permission.
Inputs	Period, closing date, reconciliation status, approval.

Main Flow

1. System checks open transactions and exceptions.

2. User reviews warnings.

3. Authorized user closes period.

4. System prevents unauthorized posting into closed period.

5. Reopening requires explicit authorized action and audit event.

Alternate Flows

1. Period can be reopened by Owner/Admin for correction according to policy.

2. Late transaction can be posted to later period with reference where policy permits.

Exceptions / Failure Handling

• Unresolved critical imbalance; unauthorized reopen; backdated posting into closed period.

Business Rules

• Closed periods are protected from ordinary edits/postings.

• Reopen must be exceptional and auditable.

Outputs

• Period status, close timestamp, user, exceptions.

Data Changes / Transaction Effects

• Accounting period state updated.

Permissions

• Close/reopen; view exceptions.

Audit Events

• Close/reopen event, warnings and approvals.

Acceptance Criteria

AC-007-01: Closed period rejects unauthorized posting.

AC-007-02: Reopen is fully audited.

UC-ACC-008 — Generate Financial Reports

Priority	Must
Primary actor(s)	Accountant, Owner/Admin, Manager
Trigger	User requests a report for a selected period.
Preconditions	Accounting data is posted and accessible; user has report permission.
Inputs	Date range, store, account/category filters and report type.

Main Flow

1. User selects report.

2. System validates date range and access scope.

3. System retrieves posted accounting data.

4. System calculates report.

5. System displays and allows export/print if supported.

6. Report includes period and filter context.

Alternate Flows

1. Reports can be generated for current/open periods with clear status.

2. Comparative periods can be selected in later analytics enhancements.

Exceptions / Failure Handling

• Incomplete configuration; unbalanced ledger; unauthorized data scope; invalid date range.

Business Rules

• Reports must use posted accounting data and identify the reporting period.

• Operational reports and financial statements must not silently use draft transactions.

Outputs

• Trial Balance, P&L, Balance Sheet, cash/bank, receivables, payables, expense reports and other configured reports.

Data Changes / Transaction Effects

• No transactional data changes.

Permissions

• View/export/print reports according to role.

Audit Events

• Report generation, export and sensitive report access.

Acceptance Criteria

AC-008-01: Trial Balance balances.

AC-008-02: P&L totals reconcile to ledger.

AC-008-03: Receivables/payables reconcile to subledgers.

AC-008-04: Report period and filters are visible.

8. Cross-Workflow Rules

• **CW-001:** A transaction must have a unique business identifier appropriate to its document type.

• **CW-002:** Source transactions and their downstream inventory/accounting effects must be linkable through stable references.

• **CW-003:** A failed downstream posting must not leave a partially committed business transaction without an explicit recoverable state.

• **CW-004:** Idempotency or duplicate-detection must be considered for integrations, especially WhatsApp and payment imports.

• **CW-005:** Historical posted documents must retain their original commercial values even when master data changes.

• **CW-006:** User permissions must be evaluated at the point of the sensitive action, not merely at screen access.

• **CW-007:** Audit records must survive normal transaction corrections and reversals.

• **CW-008:** Reports must distinguish posted/confirmed data from draft data.

• **CW-009:** Stock and accounting corrections must use controlled transactions/reversals rather than silent database edits.

9. Permissions Matrix — Baseline

Action	Admin	Manager	Accountant	Billing	Warehouse
Create/edit products	Admin	Manager	No	No	No
Change global prices	Admin	Manager	No	No	No
Customer-specific price	Admin	Manager	Limited	No	No
Create PO	Admin	Manager	No	No	No
Approve PO	Admin	Manager	No	No	No
Receive goods	Admin	Manager	No	No	Warehouse
Post invoice	Admin	Manager	Limited	Billing	No
Price override	Admin	Manager	Limited	Limited	No
Stock adjustment	Admin	Manager	No	No	Warehouse*
Customer payment	Admin	Manager	Accountant	Billing*	No

Action	Admin	Manager	Accountant	Billing	Warehouse
Supplier payment	Admin	Manager	Accountant	No	No
Expense posting	Admin	Manager	Accountant	No	No
Financial reports	Admin	Manager	Accountant	No	No
User/role administration	Admin	No	No	No	No

* Warehouse stock adjustment should be restricted to controlled, authorized adjustment types; final approval should remain with an appropriate management role.

10. End-to-End Acceptance Scenarios

• **E2E-01 Purchase:** Create supplier → create PO → partially receive 900 of 1,000 units → inventory increases by 900 → PO shows 100 outstanding → supplier invoice creates payable → partial payment reduces payable.

• **E2E-02 Retail:** Scan barcode → product found → retail price selected → GST calculated → invoice posted → payment recorded → inventory decreases by sold quantity → customer/receipt status becomes paid.

• **E2E-03 Wholesale:** Select B2B customer → customer-specific price applied → negotiated discount within permission → GST invoice posted → credit sale recorded → customer outstanding increases → later payment closes invoice.

• **E2E-04 School:** Capture school requirement → create quotation → school accepts and supplies PO → fulfil/partially deliver → invoice delivered quantity → receivable created → school payment allocated.

• **E2E-05 Accounting:** Post sales/purchase transactions + expense + customer receipt + supplier payment → generate Trial Balance → debits equal credits → P&L and Balance Sheet are generated from posted entries.

• **E2E-06 Audit:** Authorized user changes a customer price → before/after values, actor and timestamp appear in audit history; historical invoices remain unchanged.

• **E2E-07 Batch:** Receive two batches of an FMCG product → sell from valid batch → inventory and batch quantities reduce correctly → near-expiry report identifies remaining eligible batch.

11. Data Traceability

Workflow	Primary source documents	Downstream effects
Purchase	**Supplier, PO, Goods Receipt, Supplier Invoice**	**Inventory, Batch, Accounts Payable, Cash/Bank**
Retail/Wholesale	**Customer, Cart/Order, Invoice, Payment**	**Inventory, Revenue, Tax, Receivable, Cash/Bank**
School	**Requirement, Quotation, PO, Delivery, Invoice, Payment**	**Inventory, Revenue, Tax, Receivable, Cash/Bank**
Accounting	**Receipts, Payments, Expenses, Journals, Period Close**	**Cash, Bank, Receivable, Payable, P&L, Balance Sheet**

12. Requirements Deferred to Architecture/Design

• Inventory costing method (e.g., FIFO/weighted average or other chosen method).

• Chart of accounts and exact journal-posting rules.

• Invoice numbering/series strategy.

• Exact GST rule engine and current legal compliance details.

• Reservation versus physical stock issue semantics.

• Return/credit-note implementation details.

• Payment methods and reconciliation integrations.

• Offline billing strategy.

• WhatsApp API/provider and message template strategy.

• Cloud/on-premise/hybrid deployment.

• Database technology and service boundaries.

• UI framework and device/printer/scanner integration.

13. v1.1 → Design Readiness Checklist

Area	Status
Business flow	Four priority workflows are explicitly defined.
Actors	Primary roles and responsibilities are identified.
Transactions	Source documents and downstream effects are identified.
Inventory	Receipt, issue, return, batch and adjustment behavior is defined.
Pricing	Customer-specific pricing and authorization are covered.
Tax	GST data/calculation requirement is represented; exact legal implementation remains design/validation work.
Accounting	Receivables, payables, cash, bank, expenses, journals and reports are covered.
Security	Permissions and audit events are defined.
Exceptions	Common validation and failure paths are explicitly identified.
Testing	Acceptance criteria and end-to-end scenarios are provided.

End of SRS v1.1

Recommended next phase: Architecture & System Design, using SRS v1.0 + v1.1 as the requirements baseline.

SRS v1.2 — Architecture Design

Wholesale Stationery, Notebook & FMCG Business Management Platform

Baseline: SRS v1.0 + SRS v1.1 Detailed Use Case Catalogue

Business: Tumkur, Karnataka, India

Purpose: Define the domain model and component architecture before physical database schema design.

Version Relationship

This document extends SRS v1.1 and preserves its requirements baseline. It converts the v1.1 recommendation for Architecture & System Design into an architectural baseline. The four priority workflows remain the foundation: Purchase → Inventory, Retail/Wholesale Sale → Invoice → Payment → Inventory, School Requirement → Quotation → PO → Delivery → Invoice → Payment, and Accounting → Customer/Supplier/Cash/Bank/Expense → Financial Reports.

The architecture deliberately stops before physical database schema design. Database tables, keys, indexes and SQL constraints will be derived from the approved domain model in the next phase.

1. Architecture Principles

• **Modular monolith first:** one deployable application with strong internal module boundaries.

• **Domain-first:** business concepts and rules drive the architecture; the database is derived from the domain model.

• **Transaction integrity:** sales, inventory, invoicing and accounting effects must remain consistent.

• **Ledger-oriented history:** transactional history is authoritative; current balances may be optimized as projections.

• **Immutable financial history:** corrections use controlled reversals, returns, adjustments or cancellations.

• **Multi-store ready:** store/location is a first-class context.

• **Integration isolation:** WhatsApp, payment, printer and external services are accessed through adapters.

• **Analytics-ready:** historical operational data is retained for reporting and future ML/AI.

2. Domain Landscape

```
Identity & Access
Organization / Stores / Locations
Catalog — Pricing — Customers — Suppliers
Inventory — Procurement — Sales — School Business
Invoicing — Taxation — Payments — Accounting
Reporting — Analytics / AI

Cross-cutting: Audit, Notifications, Documents, Configuration, Integrations
```

3. Core Domain Model

3.1 Organization

Organization represents the business/legal context. It contains stores/locations and shared business configuration. Store/location is included in transactional context so future branch expansion does not require a structural redesign.

3.2 Catalog and Product

```
Product
■■■ Brand
■■■ Name / SKU
■■■ Category
■■■ HSN / Tax classification
■■■ Base UOM
■■■ Purchase / Sales UOM
■■■ Barcode(s)
■■■ Configurable attributes (pages, size, colour, GSM, etc.)
```

The catalogue must support both stationery/notebook and FMCG characteristics without hard-coding one category's attributes into the product core.

3.3 Units of Measure

```
Product → UOM Conversion
1 carton = 100 pieces
1 box = 12 pieces
1 piece = 1 base unit
```

The inventory engine normalizes quantities to a base unit while retaining the commercial UOM used on documents.

3.4 Inventory

```
Inventory Transaction
■■■ PURCHASE_RECEIPT
■■■ SALE / STOCK_ISSUE
■■■ SALES_RETURN
■■■ PURCHASE_RETURN
■■■ DAMAGE
■■■ EXPIRY
■■■ ADJUSTMENT
■■■ TRANSFER_IN
■■■ TRANSFER_OUT
```

Conceptual inventory position: Product + Store + Batch + Quantity. The ledger is authoritative; a current-stock projection may be maintained for performance.

3.5 Batch and Expiry

```
Product
■■■ Batch / Lot
■■■ Batch Number
```

```
■■■ Manufacturing Date
■■■ Expiry Date
■■■ Inventory Position
```

Batch-controlled FMCG must retain batch and expiry information where configured. The model also supports future FEFO/expiry-oriented stock policies.

4. Customer, School and Supplier Domains

4.1 Customer

```
Customer
■■■ Customer Type
■■■ Identity / Contact
■■■ GST Information
■■■ Billing / Shipping Address
■■■ Credit Limit / Terms
■■■ Pricing Profile
```

Customer-specific pricing and credit settings are first-class concepts. Historical invoice prices remain unchanged when customer or product master data changes.

4.2 School Business

```
School
■■■ Requirements
■■■ Quotations
■■■ School Purchase Orders
■■■ Deliveries / Fulfilment
■■■ Invoices / Payments
```

School business remains a distinct subdomain because it is periodic, bulk, negotiated and quotation/PO driven.

4.3 Supplier and Procurement

```
Supplier
↓
Purchase Order
↓
Goods Receipt
↓
Supplier Invoice
↓
Accounts Payable
↓
Supplier Payment
```

Critical rule: a purchase order is a commitment and does not increase inventory. A confirmed goods receipt posts stock.

5. Pricing Domain

```
Customer-specific price
↓
Customer-group price
↓
Quantity-based price
↓
Wholesale price
↓
Retail price
↓
MRP
```

This hierarchy is the proposed architecture baseline. Exact precedence, effective dating and authorization thresholds will be finalized during detailed design. Pricing selection must be deterministic and auditable.

6. Sales Domain

```
■■■ Retail POS
■■■ Wholesale / B2B
Sales Engine ■■■■■■■■■■ School
■■■ WhatsApp
■■■ B2B Portal
```

A common sales engine supports different channels while allowing channel-specific workflow behavior. Barcode scanning and manual product selection are both supported.

7. Invoicing and Taxation

Invoice is a posted business/financial document with an immutable or controlled historical snapshot. A dedicated taxation component calculates taxable value and GST components. The invoice stores the exact commercial and tax values used at posting.

8. Payment and Accounting Domain

8.1 Payments

```
Payment
■■■ Customer Receipt
■■■ Supplier Payment
■■■ Expense Payment
■■■ Cash
■■■ Bank
■■■ UPI / Card / Other configured methods
■■■ Allocation / Reference
```

Payments are independent transactions and can be allocated to one or more invoices. Posted payments reduce receivables/payables exactly once and are corrected through controlled reversal.

8.2 Double-Entry Accounting

```
Business Transaction
↓
Accounting Event
↓
Journal Entry
↓
Journal Lines
↓
General Ledger
↓
Financial Statements
```

The accounting domain covers receivables, payables, cash, bank, expenses, journals, period close and financial reports. Every journal must balance and source transactions must be linked to their accounting postings.

Examples

```
Sale:
Dr Customer / Cash
Cr Sales Revenue
Dr Cost of Goods Sold
Cr Inventory

Supplier purchase:
Dr Inventory
Cr Supplier Payable

Customer payment:
Dr Bank / Cash
Cr Customer Receivable
```

Exact chart of accounts and posting rules remain design-stage decisions.

9. Reporting, Analytics and AI

```
Transactional System
↓
Reporting Read Models / Queries
↓
Operational & Financial Reports
↓
Analytics Dataset / Data Pipeline
↓
Forecasting / ML
↓
AI Recommendations / Assistant
```

Reports use posted/confirmed data. The architecture preserves historical transactions so forecasting, reorder recommendations, slow-moving stock detection, customer analytics and natural-language business queries can be added later.

10. Component Architecture

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Web / Mobile UI ■
■ POS | Sales | Purchase | Inventory | Reports ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ API Layer ■
■ Controllers | Auth | DTOs | Validation ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Application Layer ■
```

```
■ Sales | Purchase | Inventory | Accounting ■
■ Customer | School | Reporting | Payments ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Domain Layer ■
■ Entities | Value Objects | Rules | Events ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Infrastructure ■
■ PostgreSQL | Storage | WhatsApp | Printer ■
■ Payment / Other External Integrations ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

11. Proposed Module Structure

```
identity
organization
catalog
pricing
inventory
procurement
sales
customers
suppliers
schools
invoicing
taxation
payments
accounting
reporting
notifications
audit
integrations
shared
```

12. Module Dependency Rules

• UI must not access the database directly.

• Business rules belong in domain/application services rather than controllers.

• Sales requests inventory and accounting effects through defined contracts rather than manipulating their storage directly.

• Cross-module references use stable identifiers and explicit contracts.

• Infrastructure implementations sit behind ports/interfaces.

• Modules may publish domain/application events rather than creating hidden side effects.

• Circular module dependencies should be avoided.

13. Transaction Boundary Architecture

```
POST SALE
■
■■■ Validate pricing / tax
■■■ Create/post invoice
■■■ Post inventory issue
■■■ Post accounting event
■■■ Commit as one consistent business transaction
```

The exact inventory posting point remains a design decision, but whichever policy is selected must prevent double stock reduction and preserve traceability between sale, invoice, inventory and accounting.

A modular monolith with a single relational transaction boundary is recommended for V1 because it simplifies atomicity across these effects.

14. Domain Events

```
SalePosted
PurchaseReceived
PaymentReceived
InvoicePosted
StockAdjusted
SalesReturnPosted
CustomerCreated
SupplierCreated
PeriodClosed
```

Initially these can be in-process events. A message broker is deferred until asynchronous scale or integration requirements justify it.

15. Integration Architecture

```
Core Platform
■
■■■ WhatsApp Adapter
■■■ Payment Adapter
■■■ Printer / Document Adapter
■■■ Email / Notification Adapter
■■■ Future External Accounting Adapter
```

The core domain remains vendor-neutral. This is especially important for WhatsApp, payment and printing integrations.

16. Security and Audit Architecture

```
Authentication
↓
User
↓
Role
↓
Permission
↓
Business Rule / Approval
↓
Transaction
↓
Audit Event
```

Backend authorization is mandatory. Sensitive actions such as price overrides, stock adjustments, invoice reversal, payments, role administration and period reopening are permission-controlled and auditable.

17. Proposed Technology Architecture

```
Frontend: Angular
Backend: Java + Spring Boot
Database: PostgreSQL
File Storage: Object/file storage
Reverse Proxy: Nginx/Caddy or equivalent
Caching: Redis only where justified
Messaging: Deferred
Analytics/ML: Separate Python/data layer later
Orchestration: Kubernetes only if operational scale requires it
```

These are architecture recommendations, not requirements lock-in. They will be validated against deployment, cost, support and operational constraints.

18. Architectural Decisions Pending Confirmation

18.1 Inventory Costing Method

Recommendation: Weighted Average Cost, unless the accounting/business requirement specifically requires FIFO or another method. This is a recommendation, not yet a locked requirement.

18.2 Accounting System Boundary

Recommendation: The platform is the primary accounting source for V1 because the SRS baseline requires full accounting. A future adapter can integrate with Tally or another external accounting system if required.

18.3 Offline POS

Decision required: **A — internet required for billing** or **B — POS continues offline and synchronizes later**. Offline POS materially affects local storage, invoice numbering, stock synchronization and conflict handling.

19. Architecture Baseline Summary

Area	Baseline
Architecture style	Modular Monolith
Domain structure	Strongly modular business domains
Inventory	Ledger-oriented; batch/expiry first-class
Sales	Unified engine for retail, wholesale and school workflows
Pricing	Dedicated deterministic pricing component
Accounting	Double-entry accounting
Transactions	Controlled posting/reversal; stable references
Security	RBAC + backend authorization + audit
Events	In-process initially
Integrations	Adapter/port based
Multi-store	Architecturally supported from V1
Analytics/AI	Separate future layer
Physical DB	PostgreSQL recommended; schema not yet designed

20. Next Architecture Phase

- **Architecture Design v0.3:** Domain Entity Model and relationship definitions.
- **ERD:** entity relationships and cardinalities, independent of SQL implementation.
- **Database Design:** PostgreSQL schema, keys, constraints, indexes and audit strategy.
- **API Design:** resource model, commands/queries, idempotency and error contracts.
- **UI/UX Architecture:** screens mapped to use cases and permissions.

Source baseline: SRS v1.1 — Detailed Use Case Catalogue. The source concludes that Architecture & System Design is the recommended next phase and lists database technology and service boundaries among the deferred design decisions.